NEUROCAD: AN AUDITABLE BOUNDED CAD COMPILER AND CONSTRAINT-EVIDENCE SYSTEM

CURRENT-EVIDENCE RESEARCH MANUSCRIPT

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22 October 2026

ABSTRACT

Computer-aided design automation requires distinctions among syntactic validity, geometric validity, conformity to stated intent, and physical adequacy. NeuroCAD is a deterministic bounded compiler and analytical-evidence system that makes these distinctions explicit. Its pipeline converts supported dimensioned text into a typed constructive-solid-geometry program, checks declared constraints, exports executable geometry, and retains source and artifact identities. A separate receipt mechanism binds analytical requests to canonical component geometry, explicit material declarations, and named loads. We consolidate the existing implementation, specifications, and retained evidence without reopening completed studies. The source-bound controlled compiler run contains 240/240 exact semantic matches, 1,000/1,000 static program checks, and 240/240 fresh kernel executions; its generated grammar and simple baselines limit interpretation. A separately frozen 216-case synthetic physics-receipt study accepted 48/48 valid requests, detected 96/96 declared contradictions or stale expected hashes, and detected 48/48 later geometry changes. A hash-only ablation detected later changes but missed all 72 semantic contradictions. One internal Linux repetition reproduced all 13 endpoint counts on the same 216 cases, without adding independent confirmation. These are positive software-contract results, not evidence of physical accuracy or general language understanding. An archival audit finds 1,211 of 1,212 listed compiler artifacts present with matching hashes; one original log is missing. We also disclose timing-digest and median-definition issues, preserve the falsified historical typed-parser claim, and distinguish later engineering extensions from confirmation. The original compiler study remains mixed; the receipt study is supported within its controlled contract. Independent language, physical-fit, and user studies remain unexecuted.

Keywords parametric CAD · deterministic compilation · provenance · conformance testing · research reproducibility

1 Introduction

A useful CAD assistant must do more than produce a plausible rendering. A request such as a dimensioned plate with a specified hole pattern concerns editable parameters, feature counts, spatial relations, and permissible failures. A script can execute while describing the wrong part. A closed mesh can omit a requested hole. A correct formula can use a dimension that no longer matches the design. These are different failure modes and require different evidence.

NeuroCAD addresses a deliberately restricted version of this problem. Its supported research method is a deterministic text-to-CAD compiler, not a trained neural model. Supported prompts are mapped to an explicit intermediate representation (IR), checked, serialized, and exported. Unsupported or incomplete requests produce errors. A separately versioned analytical layer returns calculations together with their assumptions and dependency receipts. The point of the restriction is that the admitted behavior can be specified and inspected; it does not establish competence outside that restriction.

This manuscript has two scientific objectives. First, it describes exactly what existing controlled compiler evidence demonstrates. Second, it reports a completed synthetic study of a narrow dependency-binding mechanism. The latter asks whether semantic agreement checks and geometry freshness checks detect predeclared contradictory or stale

analytical requests while accepting valid ones. The central mechanism is explicit dependency comparison: a hash tracks declared content identity, while separate numeric comparisons detect selected cross-field contradictions. Neither mechanism evaluates the physical truth of an assumption.

The repository already contains distinct scientific histories. The original typed-parser causal claim was falsified and is not rehabilitated by later compiler repairs. The compiler research line has a terminal *mixed* disposition. Physics v2 has a separate completed 216-case controlled result, now accompanied by one explicitly internal execution on another environment. Physics v3 adds engineering functionality and regression evidence, including retained failures and separately versioned repairs. This manuscript preserves these histories. It is a current-evidence synthesis and technical report, not a retrospective preregistration of a new joint experiment.

Our contributions are correspondingly bounded: (i) an explicit specification of the evaluated deterministic compiler and analytical receipt mechanisms; (ii) an evidence-qualified account of their controlled results and ablations; (iii) a claim-to-artifact map, reusable table/figure regeneration, and an archival audit that discloses missing evidence; and (iv) a precise separation between implementation correctness, measured conformance, and external validation. We claim neither a new general CAD learning algorithm nor state-of-the-art text-to-CAD performance.

2 Related work and novelty boundary

Executable and parametric CAD generation. DeepCAD models CAD construction sequences with a Transformer [12]. Text2CAD generates parametric sequences from textual instructions [4]. CAD-Recode produces executable Python CAD from point clouds [10]; its input modality differs from NeuroCAD's text interface. CAD-Coder by Guan et al. generates CadQuery programs using supervised training, geometric rewards, and planning [3]. These works already establish programmatic, editable representations and language-conditioned CAD. They are relevant prior art, not measured baselines in the retained NeuroCAD experiments.

Intent, constraints, and evaluation. ExpConCAD addresses omitted spatial constraints using construction-aware reasoning and retrieved experience [5]. CIT-CAD makes an intent tree guide both code generation and constraint verification, then uses mismatches to localize repair [2]. Thus an explicit intent representation followed by checking and repair is not, by itself, a new contribution. CADTests evaluates prompt requirements directly on generated B-reps and uses mutation-refined tests for evaluation and generation feedback [6]. HistCAD evaluates whether constraint predictions support specified dimensional edits while preserving other relations and rebuilding the complete history [1]. Initial geometry validity therefore cannot substitute for editability or requirement fidelity. PICARD provides an earlier example of constraining formal-language generation through incremental parsing [11], although its demonstrated task is SQL rather than CAD.

Provenance and geometry validity. The W3C PROV data model already represents entities, activities, derivations, and responsible agents [7]. RO-Crate describes packaging research objects with contextual metadata [9]. NeuroCAD does not claim to invent hashing, provenance, digital-thread traceability, or CAD/CAE associativity, and this package is not asserted to be RO-Crate conformant. Open CASCADE's shape analyzer checks specified geometric and topological consistency conditions [8]. Those checks are useful within their defined scope; they do not establish correspondence to a user's intended function.

Defensible comparison. The evaluated distinction is a bounded executable contract and its observed implementation behavior. The retained compiler comparators are intentionally simple reference points. Their performance cannot support superiority to Text2CAD, CAD-Coder, CIT-CAD, or an equally capable human CAD workflow. Similarly, adding a freshness mechanism to an arm that lacks one shows a mechanism contrast, not a discovery that provenance is beneficial in general. No absence-of-prior-art claim is needed for the results reported here.

3 Problem formulation and evidence levels

Let x be a prompt and L the declared supported language. The compiler is a partial deterministic map $C : \mathcal{X} \to \mathcal{P} \cup \mathcal{E}$, where $\mathcal{P}$ contains canonical programs and $\mathcal{E}$ explicit errors. A valid reference signature y records the task's declared primitive, dimensions, feature counts and sizes, and enclosure state where applicable. The benchmark success indicator is

$$s(x, y) = \mathbb{1}[C(x) = p \in \mathcal{P}]\,\mathbb{1}[V(p) = \text{valid}]\,\mathbb{1}[\sigma(p) \equiv_\epsilon y], \tag{1}$$

where V is static validation, σ the frozen signature extractor, and $\equiv_\epsilon$ its numeric comparison (absolute tolerance 10^{-6} for the controlled signature fields). Dictionary comparison requires the expected fields, rather than rejecting every extra predicted field. This is exactness over selected fields, not full geometric equivalence or complete design intent.

Table 1: Evidence levels are not interchangeable. A pass at one level does not entail the next.

Level	Observable question	Remaining uncertainty
Static admission	Does the program satisfy schema, type, graph and declared constraint checks?	Did the parser recover the intended part?
Kernel execution	Can the declared backend build a nonempty artifact?	Does it meet all dimensional and topological requirements?
Requirement conformance	Do the frozen predicates or task fields pass?	Are the predicates complete and independently correct?
Dependency freshness	Does recorded content still match its dependency?	Was that dependency authoritative or physically appropriate?
Physical/user evidence	Does a measured part or observed workflow meet its declared endpoint?	Generality beyond the measured conditions

For an invalid or unsupported prompt, fail-closed success means that no program is returned and an explicit failure is recorded. Acceptance and rejection must be reported separately: a system that rejects every request has no useful valid-input coverage. Repeated executions of one prompt are determinism observations, not new independent experimental units.

The analytical task is separately defined. A request $R = (m, x_a, c)$ consists of a versioned model m, finite explicitly named inputs x_a, and optional context c. Evaluation produces an error or

$$Y = (m_v, x_a, y_a, A, B, H_R, H_x, c_{\text{receipt}}), \tag{2}$$

where y_a are analytical outputs, A assumptions, B a claim boundary, and the hashes bind specified content. This record can be internally consistent while using an inappropriate model or an inaccurate material card.

4 Deterministic compiler specification

4.1 Language admission and representation

The evaluated compiler accepts fully dimensioned plate, box, open-top enclosure, cylinder, and sphere templates, including specified holes and slots. It parses dimension sequences using `x`, a multiplication separator, or `by`, and also named dimensions. Supported units normalize to millimetres: millimetres, centimetres, metres, and inches use factors $1, 10, 1000, 25.4$. Holes require their diameter, slots their two dimensions, and supported feature counts are bounded. Unsupported domains and missing required quantities are semantic errors. The study is not a claim that arbitrary English maps to these templates correctly.

The original canonical representation is `neurocad-ir-v1`. It contains uniquely identified primitive leaves, Boolean composition nodes, explicit roots, and optional declared constraints. Leaves include box, planar-rounded box, sphere, cylinder, cone, and torus; composition operations are union, difference, and intersection. Nodes retain translation, XYZ rotation, scale, and role. Supported constraint kinds include dimension, coincidence, offset, child count, and bounds. This CSG representation is distinct from the later experimental `neurocad-feature-ir-v0alpha1` history representation.

The latter admits feature records and semantic selectors for exact-backend experimentation. A selector names an entity kind, generating feature, predicates, optional role, and uniqueness requirement; it does not persist a raw kernel face index as its stable identity. This representation's existence does not imply that every listed feature has been validated on every backend. The original compiler results are not scores for this later representation.

4.2 Validation, construction, and serialization

The forward process is: normalize the admitted text; parse task quantities and features; construct the compatibility graph; adapt to canonical IR; validate; serialize; export; optionally execute the selected kernel; independently read the resulting artifact; then publish its verification record. Static validation uses a Draft 2020-12 schema plus semantic checks for identifier uniqueness, references, hierarchy, parameter domains, transformations, and declared constraints. The original study limits programs to 1,024 nodes and depth 128; structured input is capped at 1 MiB. These bounds limit resource exhaustion, but do not prove parser complexity for every hostile string.

Finite numeric admission must distinguish a number from a Boolean or textual number and must avoid overflow during conversion. In particular, Python's Boolean subtype relation to integers is not an acceptable admission rule. Canonical

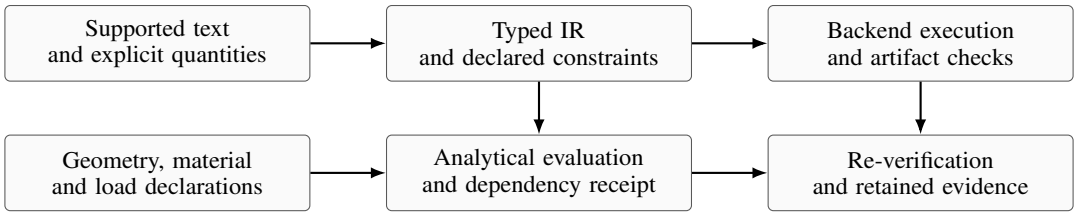


Figure 1: The compiler and receipt paths share explicit state but have separate claims. A dependency receipt does not certify a model's physical adequacy, and a kernel pass does not independently certify language understanding.

program round-trip identity is tested as $P(S(p)) = p$, with S the deterministic serializer and P the parser. Graph validation visits vertices and references, giving $O(N + E)$ structural work under bounded-size node fields; serialization and export also depend on output length. A bound on IR size does not bound the runtime of an arbitrary geometry kernel.

For the historical executed geometry study, the exporter emits OpenSCAD CSG, the kernel is OpenSCAD 2021.01, and tessellation uses `$fn=48`. Execution uses a fixed argument vector, timeout, and nonempty output checks. Retained artifacts are read through a separate mesh library and checked for the specified volume/topology/extents conditions. Mesh checks do not turn the output into exact B-rep geometry. Later STEP/BREP backends have separate engineering evidence and do not retroactively change the historical study.

4.3 Editable state and training status

A parameter edit targets a named field, followed by validation, serialization, reparse, and export. The retained edit experiment changes box width. It does not establish robust topological naming for arbitrary feature histories or designer usability. Later conversational edit and clarification behavior belongs to the current engineering implementation.

There is no learned architecture, optimizer, training loss, gradient, checkpoint, or training seed for the supported NeuroCAD method. These items are inapplicable, rather than unfinished neural components. The retrieval comparator indexes only train-labelled controlled prompts using token-set Jaccard similarity and copies the nearest reference signature. Its label access is reported explicitly.

5 Analytical receipt mechanism

5.1 Content identity and geometry scope

Canonical hashing uses sorted object keys, compact JSON separators, UTF-8 encoding, and rejection of nonfinite numbers:

$$H(z) = \mathrm{SHA256}(\mathrm{canonicalJSON}(z)). \tag{3}$$

This is content identity under one serialization contract. It is not a digital signature, proof of authorship, trust assertion, or proof that two algebraically equivalent constructions have the same geometry.

A selected component is reduced to

$$G = (\mathrm{id}, \mathrm{part}, \mathrm{operation}, \mathrm{shape}, \mathrm{parameters}, \mathrm{translation}, \mathrm{rotation}). \tag{4}$$

Display labels and project revision counters are excluded. Re-verification reloads the component and reports current when its observed $H(G)$ equals the recorded value, stale when it differs, or unavailable when resolution fails. Geometry fields excluded from G would not be protected; this makes the exact canonicalization part of the scientific mechanism.

The v2 geometry metrics are isolated-primitive closed-form quantities: volume, surface area, centroid, transformed axis-aligned bounds, and unambiguous section properties. They are not Boolean-evaluated assembly properties. For additive primitives with explicitly declared density,

$$M = \rho V_{\mathrm{mm}^3} 10^{-9}\ \mathrm{kg}. \tag{5}$$

Subtractive primitives do not receive an additive material mass.

5.2 Material and load agreement

A material card declares version, identity, revision, source, temperature where supplied, and finite properties. A load card declares version, case identity, coordinate system, source, and values. Repeated names shared with analytical

inputs must agree. The material/load condition is

$$|a - b| \le 10^{-12} \max(|a|, |b|). \tag{6}$$

The code's selected geometry checks instead use

$$|a - b| \le \max\{10^{-12} \max(|a|, |b|), 10^{-12}\}. \tag{7}$$

The absolute term is in the named quantity's numeric unit. These are software comparison tolerances, not measurement uncertainty or material tolerances.

Only three unambiguous geometry/model mappings are exercised by the v2 study. Circular-shaft torsion binds an additive cylinder's outer diameter to twice its radius and length to its height. Buoyancy compares displaced volume with the isolated additive primitive's volume. Lumped thermal response compares volume and total isolated surface area. These checks do not infer an inner bore, effective buckling length, wetted area, convection boundary subset, contact, load direction, or safety factor. In particular, the shaft formula can accept an explicitly supplied inner diameter, but this geometry binding checks only outer diameter and length; it is not proof of a matching hollow CAD solid.

5.3 Analytical models exercised in the study

Let T be applied torque in $\mathrm{N\,m}$, L, D_o, D_i dimensions in millimetres, and G_s shear modulus in $\mathrm{N/mm^2} = \mathrm{MPa}$. Circular-shaft torsion computes

$$J = \frac{\pi}{32}(D_o^4 - D_i^4), \qquad \tau_{\max} = \frac{|1000T| D_o}{2J}, \qquad \theta = \frac{1000TL}{G_s J}, \tag{8}$$

$$\sigma_{\mathrm{VM}} = \sqrt{3}\,\tau_{\max}. \tag{9}$$

Admission requires positive L, D_o, G_s, $0 \le D_i < D_o$, and finite torque. An optional supplied yield strength and safety factor define an analytical utilization. The assumptions are a straight prismatic circular shaft, linear elastic isotropic material, small twist, and uniform torque. Keyways, shoulders, fatigue, stress concentrations, and warping restraint are excluded.

For a uniform static fluid with supplied density ρ_f, gravity g, depth h in metres, and displaced volume V in cubic metres,

$$p_g = \rho_f g h, \qquad F_b = \rho_f g V, \qquad W_{\mathrm{app}} = mg - F_b \tag{10}$$

when object mass m is supplied. The model assumes complete displacement of the declared volume, excludes free-surface dynamics and stability, and does not demonstrate that a manufactured object is sealed.

For lumped thermal response, convert input volume and surface area to SI, then compute

$$L_c = V/A_s, \qquad \mathrm{Bi} = h_c L_c / k, \qquad \tau_T = \rho V c_p / (h_c A_s), \tag{11}$$

$$T(t) = T_\infty + (T_0 - T_\infty) e^{-t/\tau_T}. \tag{12}$$

Density, heat capacity, area, volume, coefficient h_c, conductivity k, and absolute temperatures must be positive; time may be zero. The evaluator rejects $\mathrm{Bi} > 0.1$. Constant properties and a uniform body temperature are assumed; radiation, phase change, contact resistance, internal heat generation, and spatial gradients are outside this model. A passing Biot check is a necessary admission rule for this implementation, not an experimentally established error bound for every object.

5.4 Conditional guarantees

Proposition 1 (Content-change detection)**.** *Assume deterministic canonicalization, a stable component read, faithful storage of the receipt, and unequal hashes for the two serialized component states. A change from G to G' is reported stale.*

Proof. The verification rule returns current only if $H(G') = H(G)$. Under the stated inequality it takes the stale branch. No property of physical geometry or author identity follows from that branch. □

Proposition 2 (Bounded contradiction rejection)**.** *If a checked repeated quantity violates its comparison inequality, all preceding admission checks complete, and the comparison is executed as specified, the full-binding evaluator rejects the request.*

Proof. The implementation raises an error on the negation of the corresponding numeric-agreement predicate, before publishing a successful receipt. □

These propositions explain why the mechanism should work and expose the limitations. Hash collisions, excluded fields, inconsistent reads, malicious receipt replacement, wrong semantic mappings, and physically false input declarations are outside their conclusions. A field that is not checked may be wrong without violating either proposition. Serialization is linear in encoded size once field order is fixed; general key sorting adds $O(\sum_j k_j \log k_j)$ comparisons for object key counts k_j. Primitive formulas are constant-size calculations. Component lookup is linear in the project size, bounded by the v2 256-component limit.

6 Experimental methodology

6.1 Controlled compiler study

We re-read the retained source-bound run `NC-REPRO-508EC40`, rather than re-executing or modifying it. Its source revision is `508ec404dd520ae16f2b4d3cf211d5b1fa46b800`; the manifest records seed 20260902, CPython 3.12.14, macOS ARM64, and OpenSCAD 2021.01. The generated 240-task dataset has 144 train, 48 validation, and 48 test labels. Families include plate, holes, slots, enclosure, box, cylinder, and sphere. Held-out labels change wording, parameters, and units under the declared grammar. They are not a random sample of independent human language.

The four comparators are an 80-mm fixed cube, raw-number extraction, unit-normalized dimensions-only extraction, and train-indexed retrieval. The last two controls help distinguish unit handling and memorization, but do not match the full feature vocabulary or inference capacity of the compiler. No learned-model parameter or training-compute comparison applies. No score against a contemporary text-to-CAD system was obtained.

The seven experiments evaluate: semantic signatures on the 240 tasks; 1,000 generated IR programs; 240 malformed cases in eight categories; 200 injected parameter corruptions with and without a declared dimension constraint; 200 box-width edits; 240 selected kernel executions; and constructed graph depths through the admission boundary. Kernel selection uses a family round-robin ordering fixed before results. The source-bound run records forced recompilation and zero resumed kernel samples.

6.2 Physics v2 frozen matrix and arms

The v2 protocol materialized 216 unique case records: 48 valid requests; 24 geometry/input contradictions; 24 material/input contradictions; 24 load/input contradictions; 24 stale expected hashes; 48 post-result geometry mutations; and 24 non-geometric mutations. Geometry-related cases cover the three mappings described above. The frozen seed is 20261011. The record is the unit of analysis; two executions of a valid record test determinism only.

The *full* arm includes semantic checks, expected-hash comparison, and post-result freshness. The *hash-only* arm retains geometry identity and freshness without the selected numeric consistency checks. The *unbound* arm evaluates the same analytical inputs without a machine-verifiable geometry/material/load dependency. Lack of an unbound freshness API is a declared mechanism difference, not an unexpected failure of a competing product.

Primary endpoints are full-arm contradiction detection over 96 cases, post-result change detection over 48 cases, non-geometric specificity over 24 cases, and valid full-arm acceptance over 48 cases. Each applicable contract requires 100% success; any miss defeats that exact contract claim. Secondary outputs include repeated scientific-field identity, report sizes, timing, and failure records. The protocol permits no exclusion merely because an arm fails.

The v2 development budget allowed 12 distinct smoke cases and at most two revisions. The first development execution failed from double resolution of a relative project path. That failure is retained. After the bounded revisions and freeze, the execution ledger records one confirmatory run. The frozen source commit and execution commit differ because the freeze artifacts were committed between them; their retained identities are listed in Appendix B. This history must not be conflated with an unrestricted search for a positive result.

6.3 Internal cross-environment repetition

On 22 October, a separately identified GitHub Actions job executed the archived v2 execution revision on Linux x86-64 with CPython 3.12.14. Its wrapper declared before execution that the primary outcomes were already known, the executing party was internal, and the attempt limit was one. The run used the frozen 216-case matrix and existing dependency lock; the recorded eight frozen-source hashes matched their originals. The original confirmatory ledger and result directory were preserved. The retained runner still emits the legacy phase name `confirmatory`; that implementation label does not convert this repetition into independent confirmation.

Table 2: Exact semantic matches in the retained generated compiler matrix. Train-labelled retrieval cases are in-sample. All columns remain controlled grammar cases. Generated directly from the per-task records.

System	Train	Validation	Test	Total
NeuroCAD	144/144	48/48	48/48	240/240
Nearest-neighbor retrieval	144/144	0/48	0/48	144/240
Unit-normalized dimensions	82/144	21/48	27/48	130/240
Raw numbers	82/144	28/48	0/48	110/240
Fixed box	0/144	0/48	0/48	0/240

The local preflight had first refused execution because available disk was below the frozen runner's 2-GiB floor. The resource gate was not lowered. The hosted environment satisfied the original resource check and produced its own declaration, execution ledger, raw outcomes, manifest, and metrics. This design tests execution portability of known synthetic outcomes under one additional environment. It introduces neither new cases nor a new physical or language-generalization endpoint.

6.4 Statistics, efficiency, and stopping

We report raw numerators and denominators. Wilson intervals accompanying historical generated matrices are descriptive; the designed cases do not justify population sampling inference. Exact paired McNemar values summarize discordances on the same fixed records. The deterministic all-cases threshold, not statistical significance, determines conformance.

Timing is host dependent and conditional. In v2, failed evaluations do not retain evaluation elapsed time, so the arms have unequal timed denominators. No latency superiority or cross-machine efficiency claim follows. Memory and runtime observations are environment diagnostics; repeated runs do not create statistical evidence about generality. The completed studies retain their original stop rules. The compiler benchmark is not rerun. The separately identified v2 repetition does not reopen the original confirmatory run. There is no new model training, physical experiment, or Kaggle execution.

7 Retained results and internal repetition

7.1 Controlled compiler behavior

Table 2 reports all five systems and all splits. NeuroCAD matches every controlled signature. Retrieval's entire 144/240 aggregate success comes from its 144 train-labelled inputs; it has 0/96 matches on the validation/test labels. Unit-normalized dimensions match 130/240 tasks, and raw-number extraction 110/240. The gains demonstrate implemented rule behavior on this contract, not independent language generalization.

Static validity, round-trip, and export checks pass for 1,000/1,000 generated programs. These 1,000 cases are not 1,000 kernel executions. All 240 malformed fixtures are rejected across the eight designed categories. In the declared constraint ablation, 200/200 injected corruptions are detected with the constraint and 0/200 without it. This demonstrates comparison against a declared equality, not discovery of the intended equality. All 200 box-width edits complete the prescribed pipeline.

The separate kernel study records 240/240 freshly compiled cases and no artifact reuse. Its volume/topology/extents checks pass within the sampled families. Depth 128 is accepted and the depth-129 case is rejected. No correctness claim extends to every geometric configuration under the size limit. The retained earlier `NC-REPRO-CB1D4A9` 999/1,000 static-program result and the historical typed-parser falsification remain part of the research history.

7.2 Physics-receipt conformance

The original frozen v2 execution completed 216 cases without recorded study execution errors. Full binding accepted 48/48 valid cases with 48/48 repeat agreement, detected 72/72 semantic contradictions and 24/24 stale expected hashes, detected 48/48 later geometry changes, and kept 24/24 non-geometric changes current. Hash-only binding detected stale hashes and later geometry changes but missed all 72 semantic contradictions. The unbound arm detected none of the later changes. These distinctions support the mechanism's specified software behavior.

For full versus unbound post-result detection, discordances are 48 versus 0, giving $2^{-47} = 7.105427357601002 \times 10^{-15}$ under the exact two-sided McNemar calculation. This small value is a fixed mechanism diagnostic, not an external

Table 3: Original Physics v2 raw-case reanalysis. Detection denominators are the predeclared families. N/A means no re-resolution API exists in that arm. Accepting a contradictory request is a failure of detection, even if its unbound analytical formula executes.

Endpoint (success criterion)	Full	Hash only	Unbound
Valid request accepted	48/48	48/48	48/48
Geometry contradiction detected	24/24	0/24	0/24
Material contradiction detected	24/24	0/24	0/24
Load contradiction detected	24/24	0/24	0/24
Stale expected hash detected	24/24	24/24	0/24
Post-result geometry drift detected	48/48	48/48	0/48
Non-geometric edit remains current	24/24	24/24	N/A

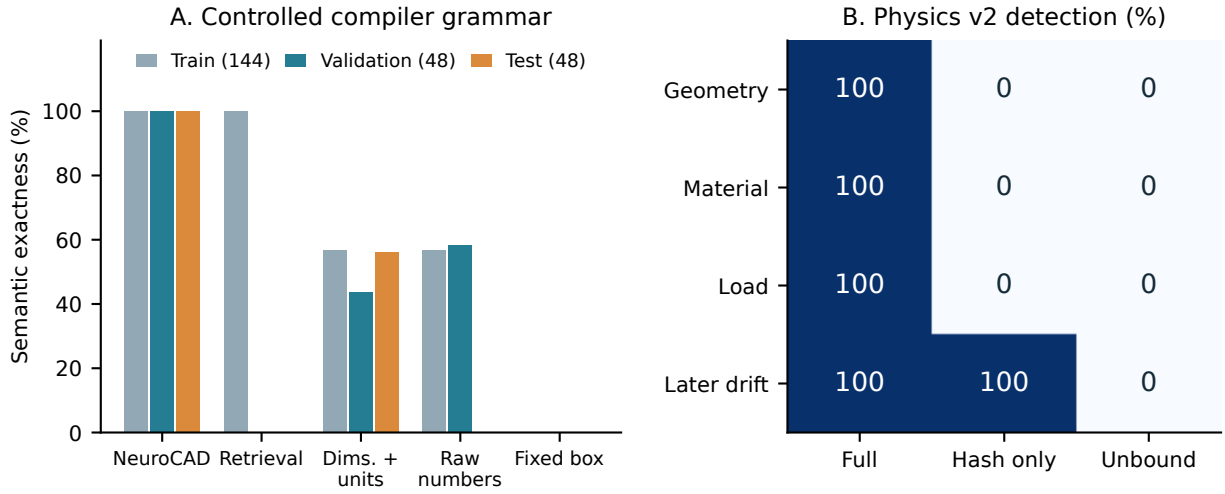


Figure 2: Retained results regenerated from raw records. Left: generated compiler task exactness, with train/validation/test separation. Right: the semantic-check versus content-freshness mechanism distinction in Physics v2. No panel represents physical validation or a new model-training result.

performance effect. The descriptive Wilson lower bounds for all-success endpoints are 86.20% for $24/24$ and 92.59% for $48/48$. They do not license claims that an unseen physical or industrial population has those success rates.

7.3 Internal repetition outcome

The hosted execution completed all 216 cases without recorded study execution errors. All 13 endpoint definitions and counts match the original metrics. For every case, the family, model, expected predicates, and applicable arm outcomes for acceptance, detection, freshness, and repeated determinism match the original raw records. The hosted ledger records `EXECUTED_ONCE`. These observations support portability of the tested contract verdicts across the two recorded environments. The raw records preserve verdicts and within-run determinism flags, not complete numerical analytical reports; cross-host bitwise equality of those numerical reports is not established. The denominator remains 216 unique designed cases; counting the repeated execution as an additional 216 independent observations would be incorrect. Raw bytes, elapsed times, and path-dependent receipt identities need not match across hosts. We make no cross-machine speed claim.

7.4 Timing and a descriptive-statistic correction

The frozen v2 report labels an upper-middle order statistic as the median for even sample sizes. To preserve its identity, we retain that report unchanged and state the convention here. Its successful-evaluation upper medians are 0.6185 ms for full binding ($n = 120$), 0.248 ms for hash-only ($n = 192$), and 0.013209 ms for unbound ($n = 216$). The conventional average-of-two middle values, re-read from the same raw records, are respectively 0.6178125, 0.2476455, and 0.0132085 ms. The frozen upper-median full-report size is 3,902 bytes; the conventional median is 3,763.5 bytes,

with maximum 4,199 bytes. This diagnostic correction does not change a primary endpoint, case outcome, or study disposition.

The uneven successful-only denominators are the more material efficiency limitation. Rejecting a contradiction omits that arm from the timed sample; the table is not a balanced cost comparison. No throughput, memory-scaling, or optimized-compute benefit is established by these medians.

8 Archival integrity and historical reconciliation

The current read-only audit verifies every available file named by the compiler manifest. Of 1,212 listed artifacts, 1,211 are present and match their SHA-256 values. The listed `logs/run.log` is missing; no mismatched present artifact is found. The numerical outcome records remain available, but complete archival preservation cannot be claimed. The missing original log is not recreated from a new execution.

The compiler's original `neurocad-deterministic-results-v1` digest also contains ten timing values. Its hash identifies the recorded bytes, but is not timing independent. A later format-v2 engineering repair separates those fields from deterministic outcomes. The historical digest and study remain unchanged. Together with the median convention above, these findings illustrate why a hash-consistent artifact can still contain a misleading measurement description.

Physics v2 has a separate retained freeze, execution ledger, raw JSONL, metrics, and generated-result source. The present audit verifies their recorded identities and confirms that regeneration of the existing result source is byte-identical. This read-only reanalysis is separate from the new hosted execution. The original environment is macOS ARM64 with Python 3.14.0; the repeated execution uses the archived source on Linux and Python 3.12.14. Neither the original run nor the internal repetition constitutes independent external reproduction. No result is represented as Kaggle evidence.

Version routing matters. Historical narratives sometimes state that a feature or release did not yet exist. Later engineering receipts can supersede a product-status statement without revising a completed scientific result. The published `v0.5.0a6` alpha and later native backend checks therefore do not change the original mixed disposition or the v2 contract boundary. The current repository state is an implementation successor, not the exact source used in every historical experiment.

9 Engineering successors and failure analysis

9.1 Physics v3 and exact geometry

Physics v3 implements temperature-aware property cards, named load combinations, cantilever mechanics, combined stress, linear thermal networks, forced vibration, bounded pipe networks, and a lumped electro-thermal model. It also provides corner enumeration, independent-uniform Monte Carlo, and local finite-difference sensitivity wrappers. These are engineering capabilities with stated assumptions, not additional v2 endpoints.

A retained v3 matrix originally passed 31/32 cases. Opposing beam loads exposed an underestimated maximum bending stress: checking only the fixed end gave zero while an interior moment extremum required 2.1 MPa. The versioned correction considers interior extrema and its known-case successor reports 37/37 source scalar cases and 52/52 installed cases or workflows. Since the failure was already observed, these are regression conformance results. They do not transform into blinded confirmation when the repaired cases pass.

Native acceptance and mesh-tagging records add backend-specific engineering evidence. The 22 October feature IR scalar repair addresses malformed inputs that could escape a structured parse error through operations such as hashing, conversion, or unchecked comparison. Of 74 new authored regression cases, the baseline failed 46 and passed 28. The repaired candidate passed 122 focused tests including existing feature/inspection coverage; a connected suite passed 158 tests with five optional-backend skips. These overlapping suites must not be summed as independent evidence. The repair improves the admission boundary; it introduces no trained model and does not change the frozen compiler or v2 scientific records.

9.2 What can still fail

Table 4: Failure analysis and evidence required to close each gap. Severity concerns the consequence for claims, not a measured incident rate.

Flaw and cause	Severity	Existing mitigation or justified change	Required additional evidence
Wrong intent passes: parser and reference share assumptions	High	Explicit supported grammar, separate signatures, error records	Independent prompt authors and adjudicators; false-acceptance analysis
Weak comparator: reduced feature vocabulary or memorization	High	Report train and held-out labels separately; limit comparative claims	Frozen grammar-independent comparator with matched task and retry budget
Stale calculation: geometry changed after evaluation	High	Content hash and re-resolution for the selected component	Broader adapters and consistent-read semantics for external systems
Wrong but current calculation: bad model or source declaration	High	Assumptions and source fields retained; no physical certificate	Qualified model selection, calibrated measurements, source governance
Partial mapping: hollow bore or boundary subset is unverified	High	List exactly which quantities were checked	Independent full-solid and boundary-condition reference checks
Numerical edge: overflow, Boolean coercion, cancellation	High	Finite strict admission and case-specific numerical regressions	Systematic domain-boundary tests and justified precision analysis
Hidden interior extremum: endpoint-only physics checks	High	Retained beam failure and separately versioned correction	Complete extremum conditions for each admitted load family
Corner envelope misses interior extrema	High	Label corner enumeration as screening	Monotonicity proof, rigorous interval method, or explicit interior search
Hash matches but archive is incomplete	Medium	Report missing log and preserve original manifest	Recovery of the original log, never a fabricated replacement
Timing diagnostic appears to prove efficiency	Medium	State successful-only denominators and median convention	Balanced instrumented successor with all attempted cases timed
Changing geometry loses edit intent	High	Named parameters and semantic selector design	History rebuild and relation-preservation benchmark; user observation
Manufactured fit or safety differs from software result	High	Physical and usability gates remain open	Calibrated physical measurements under a frozen study protocol

Table 4 makes the main remaining flaws concrete. An uncertainty wrapper deserves particular care. Evaluating all 2^d corners of a bounded box gives a global range only under additional structural conditions, such as suitable monotonicity; a nonlinear interior maximum can be missed. Independent uniform Monte Carlo samples describe the declared sampling model, not a calibrated real-world probability distribution. Finite-difference sensitivities are local and can be inaccurate near discontinuities, invalid domains, or poor step scales.

Material cards in the evaluated v2 contract are embedded declarations. Only selected geometry has a freshness re-resolution API. External material database changes and changing requirements are not automatically detected. Whole-request hashes also include path text, so moving an otherwise equivalent project may change request identity. Reproduction should distinguish that from canonical geometry-content identity.

10 Independent evaluation and falsification boundary

The existing `NC-EXT-EVAL-V0` protocol is pre-outcome methodology. It requires at least 160 independently authored prompts, including at least 120 valid and 40 reject cases, separately adjudicated references, frozen source and scorer identities, and full retention of attempted outcomes. Authors may use the public supported-scope documentation but not parser internals or hidden failure cases. A strong grammar-independent comparator is required for a comparative language-understanding claim. Valid semantic exactness and fail-closed rejection are separate primary endpoints.

That external study is unexecuted in the evidence consolidated here. Authoring more prompts after inspecting the code is useful development work, but cannot satisfy its independence requirement. A new study must respect the frozen boundary or explicitly become a different successor protocol. No physical measurements or intended-user outcomes are supplied by conformance tests, hosted CI, or the existence of a release package.

The strongest immediate falsifier for a claimed admission contract is an in-scope input whose required semantic field is wrong while the system accepts it. For freshness, it is a protected field change that remains current; for specificity, a permitted non-geometric change that becomes stale. For physical usefulness, these software counterexamples are insufficient: the falsifier must be a measured failure of the prespecified physical endpoint. The evidence needed depends on which claim is made.

11 Discussion

The positive results are useful because they identify narrow behavior that has actually been exercised. Unit normalization, feature construction, static checks, explicit dependency comparison, and artifact recording can be combined into an inspectable pipeline. The ablations show a clear mechanistic division: content identity detects changed content; semantic mapping checks detect selected inconsistencies between current content and analytical inputs. One cannot replace the other.

At the same time, complete success on a self-authored contract is partly expected. A compiler is designed to implement its grammar, and a stored hash is designed to change when the serialized content changes. The scientific value lies in an exact specification, falsifiable failure cases, fair accounting, and reproducibility, not in treating an expected conformance result as broad evidence of intelligence or physical reliability.

The repaired system can be more useful without changing a completed study's disposition. Separating successor implementation from predecessor evidence prevents hindsight from making every historical version appear correct. Keeping the wrong beam result, failed development run, missing log, and typed-parser falsification visible is necessary to understand which conclusions remain supported.

12 Limitations, ethics, and reproducibility

The central limits are author-generated data, restricted grammar and representations, simple comparators, incomplete archive preservation, absence of independent execution, and absence of physical or human evidence. This paper does not report SOTA comparisons, fabrication qualification, safety certification, or a learned-model contribution. A component-level scalar annotation is not a finite-element stress field; an analytical pipe formula is not CFD. The project should not present these different objects under interchangeable labels.

The studied synthetic records contain no personal data. The practical concern is overtrust: a reproducible and internally consistent CAD artifact can still be unsuitable for its intended use. The appropriate response is precise claim labeling and evidence appropriate to the intended application, rather than interpreting a software pass as permission to manufacture.

The numerical result tables and the result figure are regenerated by `paper/recovery_20261022/generate_evidence.py` from retained records. It writes only this manuscript's derived outputs and a source-hash audit; it launches no compiler, model, kernel, or protected study. The exact user style file is copied unchanged into this source package. Appendix B identifies authoritative records and reproduction commands. Code is in `https://github.com/THE-BU1LD/NeuroCAD` under the repository's MIT license. The scored data are repository-generated controlled fixtures; no dataset from a cited CAD learning system is evaluated here. External kernels and packaged third-party resources retain their respective licenses. This document is a complete current-evidence manuscript, not a claim that the independent external-validation program is complete.

13 Conclusion

NeuroCAD implements an inspectable bounded CAD compiler and an analytical receipt mechanism. Retained compiler evidence establishes controlled grammar and kernel conformance, while the separately frozen 216-case receipt study supports its declared semantic-consistency and geometry- freshness contract. One internal repetition reproduces its endpoint counts on another environment, while independent confirmation remains open. Present artifact auditing also identifies a missing original log and diagnostic-description issues that must accompany those results. The original study remains mixed and the historical typed-parser claim remains falsified. Broader language validity, physical fit, and user benefit require their own independent evidence; they have not been supplied by the positive software outcomes reported here.

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A Mathematical and implementation details

A.1 Why hashing and semantic checking differ

Consider a cylinder of radius r with recorded content hash h. A request can carry h correctly while specifying an outer diameter $D_o \neq 2r$. Hash-only checking passes because the component has not changed; the full arm rejects when the discrepancy exceeds Eq. (7). Conversely, changing the cylinder radius after a valid calculation invalidates the stored content hash, even if the same numeric analytical inputs remain syntactically well formed. These are different predicates over different objects.

A label-only edit is deliberately excluded from canonical geometry. That specificity prevents an irrelevant display change from invalidating the calculation's geometry identity. It does not imply semantic equivalence of every edit excluded by a future schema: exclusions need explicit review. Hash collision resistance is an assumption about content addressing; authenticating a signer would require a different mechanism.

A.2 Selected v3 equations and assumption boundaries

These formulas describe later engineering implementations and are not additional outcomes of the v2 study. For a cantilever with constant EI, tip force F, uniform load q, and length L, the linear tip response is

$$\delta(L) = \frac{FL^3}{3EI} + \frac{qL^4}{8EI}, \qquad \theta(L) = \frac{FL^2}{2EI} + \frac{qL^3}{6EI}. \tag{13}$$

With opposing signed loads, maximum absolute bending moment must also inspect admitted interior stationary points of $M(x) = F(L-x) + q(L-x)^2/2$, not only $x = 0$. The physical assumptions remain linear small-deflection Euler–Bernoulli behavior and the specified load family.

For a symmetric stress tensor $\boldsymbol{\sigma}$, principal stresses are its eigenvalues, hydrostatic stress is $\mathrm{tr}(\boldsymbol{\sigma})/3$, and

$$\sigma_{\mathrm{VM}} = \sqrt{\tfrac{1}{2}[(\sigma_x - \sigma_y)^2 + (\sigma_y - \sigma_z)^2 + (\sigma_z - \sigma_x)^2] + 3(\tau_{xy}^2 + \tau_{yz}^2 + \tau_{zx}^2)}. \tag{14}$$

This transforms a supplied local tensor; it does not solve for a spatial stress field.

A linear thermal network solves conductance balance $\sum_j g_{ij}(T_i - T_j) = Q_i$ for free nodes with supplied fixed-temperature nodes. Singular or invalid networks are rejected. For one linear forced mode, the harmonic force amplitude response is

$$|X| = \frac{|F_0|}{\sqrt{(k - m\omega^2)^2 + (c\omega)^2}}, \tag{15}$$

subject to the documented parameter and resonance checks. The electro-thermal wire model assumes $R = R_0(1+\alpha\Delta T)$ and $\Delta T = I^2 R R_{\mathrm{th}}$, yielding

$$\Delta T = \frac{I^2 R_0 R_{\mathrm{th}}}{1 - I^2 R_0 R_{\mathrm{th}} \alpha}. \tag{16}$$

The implementation rejects feedback gain at least one. Uniform temperature, constant thermal resistance, and the linear resistance law are assumptions, not outcomes verified against experiment.

The uncertainty wrappers vary at most 12 supplied scalar inputs. Corner enumeration uses 2^d evaluations. Monte Carlo uses a declared seed, independent uniform bounded draws, and 2–10,000 samples. Local sensitivity uses central finite differences around the declared base point. None automatically supplies missing physical parameter distributions.

A.3 Statistical definitions

For k successes among n fixed records, $\hat{p} = k/n$. The reported descriptive Wilson endpoints use $z = 1.9599639845$:

$$\frac{\hat{p} + z^2/(2n) \;\pm\; z\sqrt{\hat{p}(1-\hat{p})/n + z^2/(4n^2)}}{1 + z^2/n}. \tag{17}$$

For paired discordances b, c, the exact two-sided McNemar diagnostic is

$$p = \min\left\{1,\; 2\sum_{j=0}^{\min(b,c)} \binom{b+c}{j} 2^{-(b+c)}\right\}. \tag{18}$$

When $(b, c) = (48, 0)$, the result is 2^{-47}. Dependence among templates or shared failure mechanisms prevents interpreting this calculation as unqualified population evidence. No multiplicity-adjusted broad efficacy claim is made; the primary deterministic decision rule is an all-cases contract threshold.

For sorted even-n samples $u_{(1)} \le \cdots \le u_{(n)}$, the frozen diagnostic uses $u_{(n/2+1)}$, whereas the conventional median is $(u_{(n/2)} + u_{(n/2+1)})/2$. Disclosing the distinction is essential for the receipt-size diagnostic because the middle two sizes differ materially.

B Evidence map and reproduction

Table 5: Repository-relative evidence map. Paths are versioned records, not claims that every historical source matches the current checkout.

Claim or boundary	Authoritative record
Original terminal status	`TERMINAL_STATUS_2026-10-11.md`; `RESEARCH_TRUTH.md`
Compiler raw outcomes	`research/runs/NC-REPRO-508EC40/metrics/results.json`
Compiler artifact manifest	`research/runs/NC-REPRO-508EC40/manifest.json`
Compiler missing-log/digest erratum	`docs/RESEARCH_PROVENANCE_AUDIT_20261014.md`
External study boundary	`research/protocols/EXTERNAL_EVALUATION_V0.md`
Physics v2 specification	`research/physics_v2/MODEL_SPEC.md`; `research/physics_v2/PROTOCOL.md`
Physics v2 freeze	`research/physics_v2/frozen/freeze.json`
Physics v2 execution ledger	`research/physics_v2/CONFIRMATORY_EXECUTION.json`
Physics v2 raw outcomes	`research/physics_v2/runs/NC-PHYS-V2-CONFIRM-001/raw_results.jsonl`
Physics v2 analysis	`research/physics_v2/runs/NC-PHYS-V2-CONFIRM-001/metrics.json`; `research/physics_v2/audit_results.py`
V3 negative conformance	`docs/verification/physics-v3-20261014/REPORT.md`
V3 known-case repair	`docs/verification/physics-v3-corrected-20261014/REPORT.md`
Current derived tables/figure	`paper/recovery_20261022/generate_evidence.py`; `paper/recovery_20261022/evidence_sources.json`
Current separate reanalysis	`research/recovery_20261022/evidence_reanalysis.json`
Internal v2 repetition	`research/recovery_20261022/hosted-reproduction/declaration.json`; `research/recovery_20261022/hosted-reproduction/run/`
Repetition integrity audit	`research/recovery_20261022/hosted_reproduction_audit.json`
Current engineering repair	`research/recovery_20261022/engineering/`
Current full-suite execution	`research/recovery_20261022/engineering/full-tests-with-sockets.txt`; `research/recovery_20261022/engineering/full-tests.txt`
Current native demonstration	`research/recovery_20261022/native-demo/DEMO_RECEIPT.json`

Important v2 identities are retained explicitly:

- Frozen source commit: `2627c106ede6c2ac4e3570700f2bd545a763943c`.
- Execution commit: `786ba9ba5c2249a710529c3b07bf82d4e29c7f36`.
- Frozen cases SHA-256: `9caa80fb7e5257f514143ec0b297711464767c5a68ffa49988070a2f97c94c49`.
- Raw outcomes SHA-256: `97bae6a791a16088d4b9902ccc45fd0f291751d3dde822f7fad5c21c4add9023`.
- Metrics SHA-256: `34bb77ddf92225a4ee85e5c3f2a67dbca74c1cda23f60e3c9548ddf3fb013573`.

Hashes detect content differences relative to those records; they do not authenticate an execution witness. The table/figure generator records fresh hashes for every retained input it uses.

Current native demonstration. A separate 22 October engineering demonstration produced 32 hash-listed artifacts through OpenSCAD 2021.01, including plate, cylinder, and enclosure body/lid geometry. It also rejected the unsupported request to design a load-rated aircraft engine mount. The version-2 demo receipt obtains producer identity from the selected NeuroCAD CLI before and after execution, and the two identities match. Its identity scope covers on-disk project sources and packaged resources; it excludes loaded bytecode and third-party dependencies. The observed checkout revision is explicitly not treated as verification of the modified implementation. Physical validation

is recorded as not performed. These native artifact results supplement engineering reproducibility without adding compiler benchmark cases or physics-study endpoints.

The maintained current-source suite passed 2,240 tests with 12 skips and 53 reported subtest passes under Python 3.12.14 in 98.32 seconds. An initial sandboxed run retained 2,208 passes, 13 skips, four failures, and 27 errors; all 31 failures/errors were socket-permission failures. The successful rerun enabled local test-server sockets without weakening code or tests. Both logs are retained under the engineering evidence directory. Suite, subtest, and focused-regression counts overlap and are not independent scientific sample sizes.

Regenerate this manuscript only. From the repository root:

```
python paper/recovery_20261022/generate_evidence.py
cd paper/recovery_20261022
pdflatex -interaction=nonstopmode -halt-on-error main.tex
bibtex main
pdflatex -interaction=nonstopmode -halt-on-error main.tex
pdflatex -interaction=nonstopmode -halt-on-error main.tex
```

Python requires Matplotlib and NumPy for the derived figure. LaTeX requires the included unchanged `arxiv.sty` and the packages declared in `main.tex`. PDF compilation is separate from scientific execution.

Read-only compiler archive check.

```
python scripts/audit_retained_artifacts.py \
  research/runs/NC-REPRO-508EC40/manifest.json
```

The presently missing original log means this audit returns failure for full archival completeness. That is the expected disclosure, not a command to repair or overwrite the archive.

Scientific reproduction. The original compiler uses its source-pinned reproduction procedure and native OpenSCAD dependency. Physics v2's independent reproduction instructions are in `research/physics_v2/EXTERNAL_REPRODUCTION.md`; its completed single-attempt confirmatory directory must not be reused. A legitimate independent reproduction is a separately identified run with its own environment and output records. It does not mutate the original ledger or become a second attempt to rescue the original hypothesis. In the present session, the local v2 preflight was blocked by the frozen runner's 2-GiB free-disk requirement; it did not launch the cases locally. The subsequent internal hosted repetition is separately recorded at `https://github.com/THE-BU1LD/NeuroCAD/actions/runs/39726824693`. Its workflow revision is `95aa315e87e0546191bbcdbce60f9c083b43b4b6`; the evaluated source remains `786ba9ba5c2249a710529c3b07bf82d4e29c7f36`. The retrieved artifact archive has SHA-256 `4c8c9e7af18fcaed4a88d3ecd0bfa00d9b5161ae898ee20c7840d17e2220d217`. The frozen JSONL file hash and the runner's canonical parsed-case digest refer to different byte representations and must not be compared as if they were the same hashing contract.

Release boundary. This manuscript and its source are a new synthesis package. No predecessor paper, frozen protocol, dataset, run directory, or release tag is replaced. No arXiv submission, external review, or independent participant contribution is implied by compiling a PDF.